

Anil Kumar

AI/ML Engineer – Computer Vision, Multimodal
AI & Image/Video Understanding

☎* +91 98841 80067
✉[regular] anilcs13m@gmail.com
🌐 anilcs13m.github.io
in anilcs13m

Summary

ML/AI Engineer with 9+ years building and operating **computer vision and multimodal systems in production**. I own the full ML lifecycle—data strategy, model architecture, training, low-latency inference, monitoring, and cost optimization—and I have built systems for video understanding, content safety, recommendation, visual search, and monetization. Known for turning applied research into **reliable products with measurable business impact**.

Education

M.Tech in CS	Indian Institute Of Technology–Madras	2013–2015
B.Tech in CS	Gautam Buddha Technical University-Lucknow	2007–2011

Professional Experience

Nov 2020 – **ZEE Entertainment Enterprises – Software Engineer III**

Present Technical lead for the computer vision and multimodal AI platforms behind video understanding, safety, monetization, and creative automation across the streaming catalog, working alongside product and infrastructure teams to take models from prototype to monitored production.

- **Video Moderation Platform:** Built a production safety pipeline that fuses temporal video modeling, object detection, OCR, aesthetic scoring, similarity search, and scene segmentation to flag unsafe and policy-violating content. **Impact:** automated 90% of unsafe-content filtering and cut manual review effort by 60% while holding to strict latency and reliability targets.
- **Cold-Start Engagement Ranking:** Designed a cost-aware multimodal engagement model that combines visual and audio representations with adaptive temporal sampling for short- and long-form video. **Impact:** improved ranking quality for newly published videos and reduced inference cost through selective, content-aware model routing.
- **Thumbnail Optimization System:** Created an automated thumbnail-selection pipeline using face detection, emotion recognition, and aesthetic scoring to surface the most compelling frames. **Impact:** Increased click-through rates across content feeds and creator profiles.
- **Scene-Level Video Understanding for OTT:** Led the architecture that converts long-form video into structured metadata via face recognition, actor identification, person tracking, object detection, vector indexing, video-to-text, and LLM summarization, powering semantic search, X-Ray actor lookup, scene navigation, and contextual ad placement.
- **Personalized Genre-Discovery Rails:** Built a recommendation stack with Two-Tower retrieval, deep ranking, and sequential re-ranking over behavioral, content, and engagement signals to personalize genre rails for millions of users. **Impact:** improved NDCG ranking by 23%.

- Feb 2018 – **Charmboard (TAGOS) — Software Engineer, ML / Computer Vision**
- Sept 2020 Designed and shipped **shoppable-video and visual-commerce systems** that let viewers discover and buy fashion items directly from video—an early “*Shop the Look*” experience for creator-led commerce.
- **Visual & Multimodal Search:** Built image- and text-based fashion search on learned visual embeddings and vector indexing, letting users query products from video frames or natural language. **Impact:** achieved 90% Recall@5 and 95% Recall@10, improved relevance of discovery and reduced dependence on manual catalog tagging.
 - **Shoppable Video Injection (Shop the Look):** Designed a video-to-commerce pipeline that detects fashion items in video and maps them to purchasable products using visual similarity search, object detection, and person tracking. **Impact:** enabled seamless in-stream shopping without disrupting playback and removed most manual product tagging.
 - **Data-Centric Fashion AI:** Curated domain-specific fashion datasets (images, captions, metadata) and trained models for classification, captioning, and video analysis to power the shoppable-video and creator-commerce workflows.
- Jan 2017 – **AllGoVision – Software Engineer**
- Oct 2017 Developed and tuned **deep-learning face recognition systems**, owning data curation, metric learning, embedding refinement, and production threshold calibration.
- **Face Recognition System:** Trained metric-learning models on a curated set of 3.4K identities (~1K images each) and scaled embedding experiments to 10K identities. **Impact:** achieved 98.2% LFW benchmark accuracy and improved target-demographic recognition accuracy from 86% to 94% through targeted data curation, embedding optimization, and calibrated decision thresholds.
- Oct 2016 – **AllGoVision – Software Engineer Intern**
- Dec 2016 Built **computer vision data pipelines** supporting face recognition and OCR/ANPR model development for real-world visual analytics.
- Developed preprocessing workflows for augmentation, outlier removal, class balancing, and similarity-based filtering.

Technical Skills

- CV& Video** Video understanding, object detection, face recognition, person tracking, visual similarity search, scene understanding, content safety/NSFW moderation.
- Multimodal & LLM** Image/video-text retrieval, multimodal embedding alignment, Temporal modeling, RAG, LLM integration, agentic workflows.
- ML Systems** ML system design (HLD/LLD), scalable training/inference architecture, GPU inference optimization, Model monitoring, Batch inference, latency/cost trade-offs.
- Core ML** Metric & representation learning, ranking and retrieval, deep ranking, model fine-tuning, model evaluations, data-centric ML.
- Tools** Python, PyTorch, TensorFlow, OpenCV, FAISS, FastAPI, Docker, AWS, GCP, vector databases.

Datasets & Data Creation

Owned creation of **large-scale domain-specific datasets** for production ML systems, especially in areas where public datasets were unavailable or not representative of the target domain.

- **Face Recognition:** Curated 3.4K–10K identity datasets for metric learning, embedding optimization.
- **Fashion AI:** assembled e-commerce image corpora for visual similarity search and shoppable video.
- **Fashion Object Detection:** built Indian-apparel detection datasets for categories underrepresented in public benchmarks.
- **Video Moderation:** curated short-form video datasets for NSFW, unsafe content, and watermark/logo detection.